

ADVANCE PROBLEM PROGRAMMING

SLO 1: Bohm-Jacopini Theorem

1. Identify and implement the three control structures in Java.

The Bohm-Jacopini Theorem states that any program can be written using three basic control structures:

a) Sequence: Statements are executed one after another.

```
public class Main {  
    public static void main(String[] args) {  
        int a = 10;  
        int b = 20;  
        int sum = a + b;  
  
        System.out.println("Sum = " + sum);  
    }  
}
```

b) Selection: A decision is made based on a condition using if, if-else, or switch.

```
public class Main {  
    public static void main(String[] args) {  
        int marks = 75;  
  
        if (marks >= 50) {  
            System.out.println("Pass");  
        } else {  
            System.out.println("Fail");  
        }  
    }  
}
```

c) Iteration: A set of statements is repeatedly executed using loops such as for, while, or do-while.

```
public class Main {  
    public static void main(String[] args) {  
        for (int i = 1; i <= 5; i++) {  
            System.out.println(i);  
        }  
    }  
}
```

Thus, sequence, selection, and iteration are the three fundamental control structures of structured programming.

2. Convert a Java goto-like scenario (nested if-break) into structured code.

Java does not support the goto statement. A nested if with break can be replaced with a structured loop and condition.

Goto-like approach using nested if-break:

```
public class Main {  
    public static void main(String[] args) {  
        for (int i = 1; i <= 10; i++) {  
            if (i == 5) {  
                break;  
            }  
            System.out.println(i);  
        }  
    }  
}
```

Structured version:

```
public class Main {  
    public static void main(String[] args) {  
        int i = 1;  
        while (i < 5) {  
            System.out.println(i);  
            i++;  
        }  
    }  
}
```

Output:

```
1  
2  
3  
4
```

The structured version avoids unnecessary jumping and clearly defines the condition under which the loop continues.

3. Explain the importance of structure in improving program readability.

Structure is important because it makes a program clear, organized, and easy to understand. Structured programming uses sequence, selection, and iteration instead of uncontrolled jumps.

The main advantages are:

Improves readability – The flow of the program is easy to follow.

Simplifies debugging – Errors can be located more easily.

Improves maintenance – Changes can be made without affecting the entire program.

Reduces complexity – Unnecessary jumps and confusing control flow are avoided.

Improves reliability – Programs become easier to test and understand.

Promotes modular programming – Large programs can be divided into smaller, manageable sections.

Therefore, the Bohm-Jacopini approach helps create programs that are structured, readable, maintainable, and reliable.